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CarbDisMut: database on neutral and disease-causing mutations in human carbohydrate-binding proteins

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Protein-carbohydrate interactions are involved in several cellular and biological functions. Integrating structure and function of carbohydrate-binding proteins with disease-causing mutations help to understand the molecular basis of diseases. Although databases are available for protein-carbohydrate complexes based on structure, binding affinity and function, no specific database for mutations in human carbohydrate-binding proteins is reported in the literature. We have developed a novel database, CarbDisMut, a comprehensive integrated resource for disease-causing mutations with sequence and structural features. It has 1.17 million disease-associated mutations and 38,636 neutral mutations from 7,187 human carbohydrate-binding proteins. The database is freely available at https://web.iitm.ac.in/bioinfo2/carbdismut. The web-site is implemented using HTML, PHP and JavaScript and supports recent versions of all major browsers, such as Firefox, Chrome and Opera.

Key words: carbohydrate-binding proteins; database; disease-causing; mutations; neutral.

Introduction

Proteins interacting or covalently bound to carbohydrates are known as carbohydrate-binding proteins such as lectins, carbohydrate binding modules, glycoproteins and glycoconjugates. Carbohydrate-binding proteins play an important role in cell recognition, adhesion, folding and developmental process. Mutations in amino acid residues of carbohydrate-binding proteins affect binding affinity, resulting in defective cellular and biological functions (Ray et al. 1979; Shanmugam et al. 2022). Su et al. (2010) studied the effect of mutations in CBM16 and found that the Q121E mutation enhanced substrate affinity, whereas the Q21G and N97R mutations decreased substrate affinity.

Mutations can be neutral or cause diseases such as cancer, neurological, immunological, and respiratory. Larsen et al. (2004) observed structural changes in mannose-binding lectin (MBL) protein due to mutations that reduced the potential to stimulate immunological responses. Further, alteration in MSH2 gene leads to H839R mutation, which causes colorectal cancer (Sakai et al. 2016). Parkinson disease and large intestine cancer are linked to podocalyxin-like (PODXL) gene (R294Q) (Sudhaman et al. 2016; Pignani et al. 2018).

The information on disease-causing and neutral mutations is available in databases such as Humsavar (http://www.uniprot.org/docs/Humsavar), SwissVar (Mottaz et al. 2010), ClinVar (Landrum et al. 2014), 1,000 Genomes (Auton et al. 2015) and COSMIC (Forbes et al. 2015). On the other hand, several databases have been reported for the structures, affinities and functions of protein-carbohydrate complexes (Shanmugam et al. 2020; Veluraja et al. 2020). Specifically, ProCarbDB (Copoiu et al. 2020) and UniLectin3D (Bonnardel et al. 2019) provide data for three-dimensional structures, GlyGen is an integrated database for carbohydrate and glycoconjugate related data (York et al. 2020), ProCaff is a collection of data on binding affinity of protein-carbohydrate complexes (Shanmugam et al. 2020) and CAZy accumulates the enzymatic functions of carbohydrate-active enzymes (Cantarel et al. 2009). However, there is no database available for disease-causing mutations in carbohydrate binding proteins.

Sequence and structural features are generally used to identify disease causing mutations (Pandey et al. 2022). In our earlier work, we have developed algorithms for predicting the binding affinity of protein-carbohydrate complexes and their mutants (Shanmugam et al. 2021, 2022). In this work, we report a database for disease-causing mutations in carbohydrate-binding proteins, CarbDisMut, which includes information on carbohydrate-binding proteins and mutational effects at four levels: gene, protein, carbohydrate, and diseases. It would serve as a useful resource for understanding sequence-structure-function-disease relationship in carbohydrate binding proteins.

Contents of the database

We have collected the information on carbohydrate-binding proteins and protein-carbohydrate complexes from various resources such as ProCaff (Shanmugam et al. 2020), UniProt (UniProt Consortium 2021), GlyConnect (Alocci et al. 2018), GlyGen (York et al. 2020) and PolarProtDB (Zeke et al. 2021) using "keywords" search. ProCaff, GlyConnect, GlyGen and PolarProtDb are well maintained databases for carbohydrate-binding proteins, which provide reliable information on experimental binding affinity of protein-carbohydrate complexes, glycosylated and covalently linked carbohydrates,

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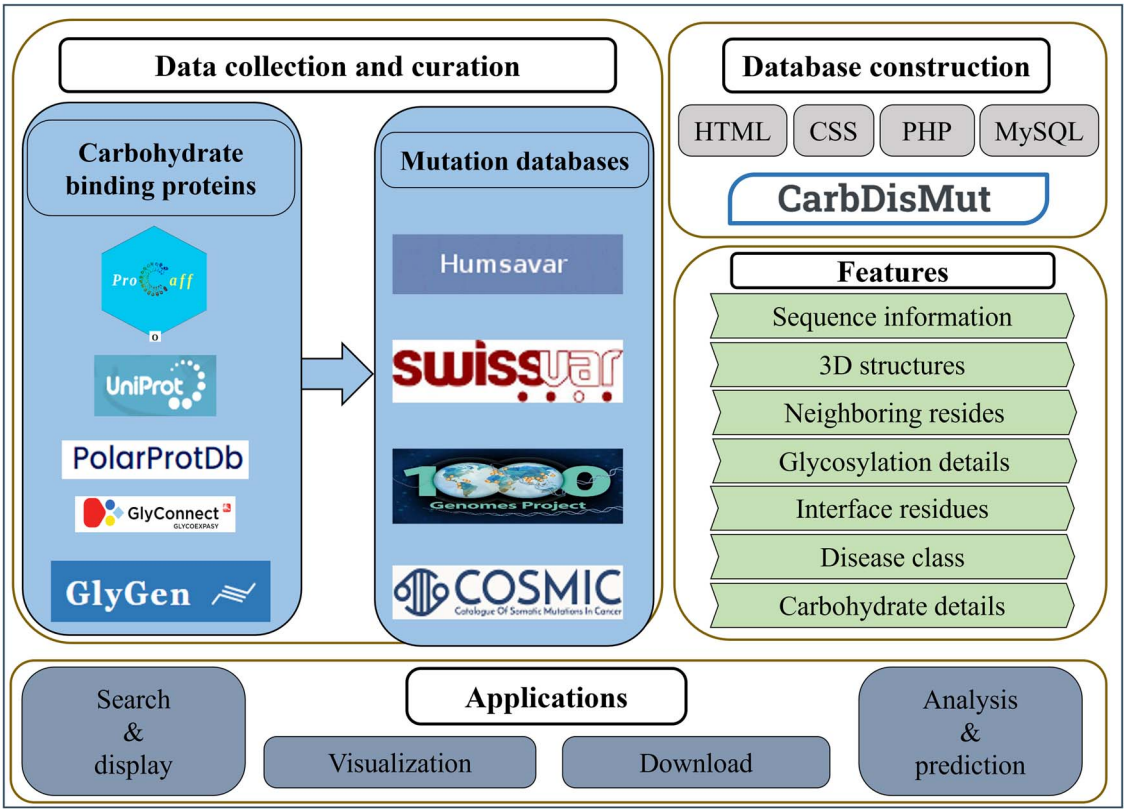


Fig. 1. Schematic diagram describing data collection, features and applications of CarbDisMut. The resources used to collect data on carbohydrate binding proteins as well as disease-causing and neutral mutations are mentioned in the "Data collection and curation" box. We have used HTML, CSS, PHP and MySQL as a database system to construct the CarbDisMut database; (ii) the database is stored in a SQL table with column names and PHP is being used to fetch them and display in the front-end using HTML.

type of glycosylation evidence and glycosylation during post translational modifications, respectively. We have used keywords such as names of carbohydrate/sugar/glycan binding proteins, lectins, carbohydrate binding modules, and carbohydrates, which includes commonly occurring monosaccharides (glucose, fructose, fucose, glucosamine, sialic acid), disaccharides (lactose) and polysaccharides (cellulose, glycogen, starch). Further, we have searched the UniProt database using the above-mentioned keywords and are manually filtered based on annotation details given in it. In addition, we have filtered and selected the proteins only from human, which are interacting with carbohydrates based on the annotations from UniProt. These criteria yielded a total of 7,187 proteins.

We followed the same methodology described in MutHTP (Kulandaisamy et al. 2018) for curating the database. Initially, we have downloaded all the mutations from well-known variant databases such as 1,000 Genomes (Auton et al. 2015), ClinVar (Landrum et al. 2014), COSMIC (Forbes et al. 2015), SwissVar (Mottaz et al. 2010) and Humsavar, and extracted the mutations in human carbohydrate binding proteins.

Humsavar and SwissVar manually classified disease and neutral mutations based on the information available in the literature. 1,000 Genomes data was derived from UniProt human variants file and includes mutation origin, disease-related mutation, and UniProt accession number. ClinVar evaluates the clinical significance of mutations detected in patient samples provided by different submitters. Pathogenic/-likely pathogenic and benign/likely benign are classified as disease and neutral mutations, respectively, in 1,000 Genomes and ClinVar. In addition, benign entries with disease names are considered as disease-causing mutations.

COSMIC is a cancer-specific manually curated database with genome-wide screening data. We considered disease-causing mutations based on occurrence, i.e. if a mutation occurs more than once in a cancer gene, it is treated as disease mutation (Gnad et al. 2013). COSMIC lists gene name, gene bank accession number, and ensembl transcript/genome id at gene level. We mapped the ensembl transcript id and gene name against UniProt to get the UniProt accession number, which annotates at protein level.

In Hamsavar, SwissVar, 1,000 Genomes and COSMIC databases, the same disease is denoted with different names and hence, we manually checked, curated, and divided them into 14 classes based on KEGG Diseases (Altermann and Klaenhammer 2005). We have used our in-house Perl scripts for data curation. During the data preprocessing step, each protein mutation was mapped with its respective UniProt identifiers using the gene name, transcript and RefSeq id. Next, we considered mutations (insertions, deletions, missense and nonsense) that are associated with the carbohydrate-binding proteins. The detailed workflow, features and applications of the database are shown in Fig. 1. Further, we have used our earlier method (PCA-MutPred) to predict the free energy change upon mutations ($\Delta\Delta G$ values), where

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Table 1. The description of data items in CarbDisMut with an example entry showing the data of N97S mutation in ITGB1 along with other information.

Entry ID	**40610**
Gene level information	
Gene Name	ITGB1
Nucleotide Mutation	c.290A>G
Mutation type	Missense
Chromosome number and Genome position	10:32929908-32929908
Origin	Somatic
Protein level information	
UniProt ID	P05556
Mutation	N97S
PDB, position & chain ID	3VI3_N77:B,N77:D 3VI4_N77:B,N77:D 4WJK_N77:B 4WK0_N77:B 4WK2_N77:B 4WK4_N77:B
PDB ID	3VI3 | 3VI4 | 4WJK | 4WK0 | 4WK2 | 4WK4
Experimental 3D structure	Known
Interface residues	3VI3, 3VI4, 4WJK, 4WK0, 4WK2, 4WK4
Predicted ΔΔG	1.413
Neighboring residues	NVTNRSK
Conservation Score	0.814
Carbohydrate level information	
Glycosciences.de	3VI3 | 3VI4 | 4WJK | 4WK0 | 4WK2 | 4WK4
GlyGen	P05556
Glycosylation type	N-linked ITB1_HUMAN
GlyTouCAN ID	G42666HT, G55220VL, G09724ZC
Disease level information	
Disease name (COSMIC: Tissue type is given)	Prostate
Disease class	Cancers
Source	COSMIC
Clinvar ID	NA
Cosmic ID	COSM240258
Humsavar ID	NA

Hyperlinks to other databases are underlined

three-dimensional structures are available (Shanmugam et al. 2022).

The database contains information on four levels that include, (i) *gene*: name, chromosome and genome position, nucleotide mutation, type of mutation, and their origin like somatic or germline, (ii) *protein*: sequence and 3D structure, mutation details, neighboring residues, conservation scores, protein-carbohydrate interface residues and predicted free energy change upon mutations, (iii) *carbohydrate*: structural information from pdb2linucs, pdb-care and CARP (Böhm et al. 2019) and glycosylation type, and (iv) *diseases*: name of the disease associated with each mutation and its classification based on KEGG human diseases information. Each data is cross-linked with major databases such as UniProt, PDB, GlyGen, Glycosciences.de (Böhm et al. 2019), dbPTM (Li et al. 2022), ClinVar and COSMIC. Further, the glycosylation type of each mutation is obtained from protein post-translational modification information database, dbPTM and cross-linked with available mutations. An example data for gene ACE2 and mutation G764R is given in Table 1. The location of the mutation site is visualized using the JSMol applet.

Data retrieval

Users can retrieve data from CarbDisMut, which offers a variety of search and display options. An example of data retrieval from CarbDisMut is illustrated in Fig. 2. In this example, we build a query using a combination of multiple search options such as mutation type "missense", experimental 3D structure "Known", disease class "Cancers", glycosylation type "O-Linked" and predicted ΔΔG "between -14 and 7" (Fig. 2A). In the second step, users have the option to select the columns to be displayed (Fig. 2B). After submitting the query, the results are displayed in a table format (Fig. 2C). On the result page, each entry ID has a hyperlink for their respective individual page, which contains the details of the data along with structural visualization. The search results can also be downloaded in a tab-separated file for further analysis.

Unique features

The existing carbohydrate related databases are mainly focused on binding affinity, structural and functional information. CarbDisMut is the first database for the mutations of human carbohydrate-binding proteins cross-linked with sequence, structural and mutational information on gene, protein and carbohydrate levels. Specifically, each entry has information on mutation, experimentally determined structures, mutations mapped with structures, conservation scores, carbohydrate information in LINUCS and PDB formats, glycosylation details only from dbPTM, disease-associated information and source of the mutation. Further,

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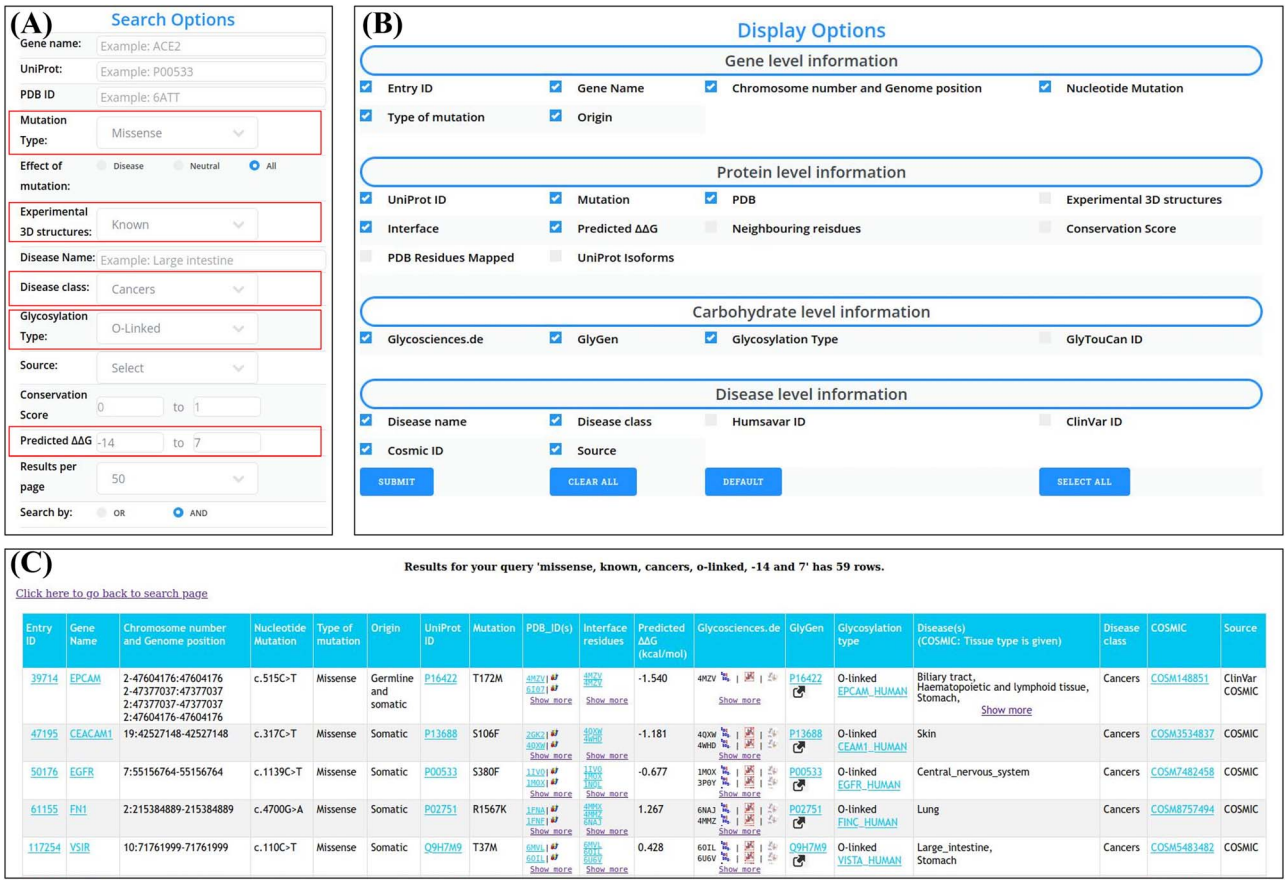


Fig. 2. An example of data retrieval from CarbDisMut database: (A) search options, (B) display options, and (C) results.

options are available to visualize the protein, carbohydrate, mutation site as well as interface residues with a cutoff of ≤5 Å. An example for JSmol visualization using PDB ID 1A7S (Entry ID: 942302) is shown in Supplementary Fig. S3. In addition, CarbDisMut is cross-linked with protein, carbohydrate, functional and mutation databases.

Database statistics

CarbDisMut contains 1.17 million disease-causing mutations and 38,636 neutral mutations from 7,187 unique carbohydrate-binding proteins. We have predicted ΔΔG values for 76,622 (72.6%) and 3,728 (86.9%) mutations at the interface on disease and neutral classifications, respectively, and the data are included in the database. Additional details on statistics is provided in supplementary information.

Applications

The comprehensive integrated CarbDisMut database facilitates to provide data for a mutation along with sequence and structural information of protein and carbohydrate. It has option to view and download the data. The data will be useful to understand the features influencing disease-causing mutations using sequence and structural features. In addition, the dataset can be used to develop computational tools for discriminating disease-associated and neutral mutations. It will also be helpful to relate the change in binding affinity upon mutation with neutral/disease-causing mutations.

For example, human alpha-L-iduronidase (UniProt ID: P35475) complexed with iduronic acid (PDB ID: 3W82) has a disease-causing mutation, G84R. The substrate, iduronic acid is a monosaccharide and major uranic acid component of the glycosaminoglycans (GAGs). Further, the mutation, G84R in this complex leads to a mucopolysaccharidosis, which belongs to the disease class of congenital disorders of metabolism. We have predicted the free energy change upon mutation (ΔΔG) using our method PCA-MutPred and the result showed that it decreased the binding affinity of 2.862 kcal/mol.

Further, we have validated the change in binding affinity and stability of G84R in alpha-L-iduronidase-iduronic acid complex using different methods and explored the structural changes upon mutation. We have constructed the mutant, G84R using FoldX and obtained the mutated alpha-L-iduronidase -iduronic acid complex structure using AutoDock4, which is visualized using the PLIP webserver (https://plip-tool.biotec.tu-dresden.de/plip-web/plip/index). The analysis on comparison of structures showed that due the mutation, interaction of alpha-L-iduronidase with iduronic acid lost three hydrogen bonds and reduced the strength of salt bridges. The wild-type (G84) structure has 10 hydrogen bonds and 3 salt bridges whereas mutated (R84) has 7 hydrogen bonds and the distance between charged atoms, which are involved in salt bridges are higher in the mutant structure compared to wild-type. The comparison of wild-type and mutation structures are presented in Supplementary Fig. S4.

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The decrease in binding affinity upon mutation has been verified with two other methods: (i) we used our server, PCAPred (structure-based) to predict the binding affinity (ΔG) of wild type (G84) and mutant (R84) complexed with iduronic acid. The binding affinities are −9.05 and −8.33, respectively, which showed that the mutation in the complex reduced the binding affinity by 0.7 kcal/mol and (ii) we used a structure-based stability prediction webserver DDMut (https://biosig.lab.uq.edu.au/ddmut/) to examine the stability of a protein upon mutation. We observed that the mutation G84R decreased the stability of −1.79 kcal/mol.

The applications of PCA-MutPred on carbohydrate binding effects have been illustrated with several other complexes and three of them are discussed below: (i) human protein O-mannose beta-1,2-N-acetylglucosaminyltransferase (Q8WZA1) in complex with Mannosyl-peptide (PDB ID: 5GGI) has disease causing mutation, R488Q. It leads to a muscular dystrophy-dystroglycanopathy type B3 (MDDGB3), which belongs to musculoskeletal diseases and nervous system diseases classes. We have predicted the free energy change upon mutation ($\Delta\Delta G$) using our method PCA-MutPred and the result showed that it decreased the binding affinity of 1.682 kcal/mol, (ii) Hemoglobin subunit beta (P68871) complexed with alpha-L-fucopyranose-(1-6)-2-acetamido-2-deoxy-beta-D-glucopyranose (PDB ID: 4X0L) has a disease-causing mutation, L92Q, which leads to lung cancer (Adenocarcinoma). We have predicted the free energy change upon mutation ($\Delta\Delta G$) using PCA-MutPred, which showed a decrease in binding affinity of 1.90 kcal/mol and (iii) P52S mutation in the same complex leads to a skin cancer, which belongs to a malignant melanoma. It decreased the binding free energy ($\Delta\Delta G$) of 2.01 kcal/mol using PCA-MutPred. Hence, combining free energy change with disease causing and neutral mutations could facilitate to progress on glycoinformatics research.

Implementation and future update

The website is implemented using HTML, PHP and JavaScript and supports recent versions of all major browsers, such as Firefox, Chrome and Opera. The database will be updated periodically and the updated information will be reflected in the What's new section of the database.

Author contributions

N.R. Siva Shanmugam (Data curation [equal], Formal analysis [equal], Investigation [equal], Methodology [equal], Resources [equal], Writing—original draft [equal]), A. Kulandaisamy (Data curation [equal], Formal analysis [equal], Investigation [equal], Resources [equal], Writing—review & editing [equal]), K. Veluraja (Conceptualization [equal], Methodology [equal], Writing—review & editing [equal]), and Gromiha Michael (Conceptualization [equal], Funding acquisition [equal], Methodology [equal], Project administration [equal], Supervision [equal], Writing—review & editing [equal]).

Supplementary material

Supplementary material is available at *Glycobiology Journal* online.

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Conflict of interest statement

None declared.

Data availability

The database is freely available at https://web.iitm.ac.in/bioinfo2/carbdismut.

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